AgaveChem: A Neural Atom-to-Atom Mapper Trained on Expert Template and Maximum Common Substructure Ground Truth

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Abstract

Atom-to-atom mapping (AAM)—the assignment of correspondence between reactant and product atoms in a reaction SMILES—is a prerequisite for reaction template extraction, retrosynthetic analysis, reaction classification, and the training of synthesis prediction models. Computational approaches to AAM broadly divide into graph-based substructure-matching algorithms, which are interpretable but scale poorly to complex reactions; template-based methods, which apply curated reaction rules to establish atom correspondence within known reaction classes but are limited by the coverage of their rule base; and neural mappers, typically trained via unsupervised methods on large corpora of unlabeled reaction SMILES. While neural methods can achieve strong overall accuracy, an unsupervised objective has no mechanism to explicitly penalize incorrect correspondences, leading to systematic errors on reactions with high symmetry or where multiple plausible assignments exist. Recent work has shown that supervised training from manually labeled reaction data improves accuracy, but hand annotation of individual reactions scales poorly—even ambitious human-in-the-loop efforts cover only thousands of reactions.

We present AgaveChem, an open-source Python library whose primary contribution is a supervised neural atom-to-atom mapper. Rather than annotating individual reactions, we curate a library of reaction SMIRKS templates and compose them with a maximum-common-substructure (MCS) mapper to generate atom-mapping ground truth from a filtered subset of the Lowe USPTO dataset, without any per-reaction manual annotation. A single verified template propagates correct supervision to every reaction in its scope, yielding a labeled training corpus orders of magnitude larger than what direct annotation can realistically provide. The resulting labeled reactions—fully mapping ∼60% of the corpus and covering ∼90% of product atoms—supervise an ALBERT-based neural mapper through a two-phase training procedure: unsupervised MLM pre-training followed by multitask fine-tuning that simultaneously optimizes MLM and a direct attention alignment objective against the generated ground truth mappings. On held-out benchmark reactions, AgaveChem achieves greater accuracy than existing neural mappers. The deterministic mappers developed to generate the training labels—an identical-fragment mapper, an expert template mapper, and an MCS mapper—are released as independent, composable library components. As part of this work we also release `rdchiral_plus`, an improved fork of the rdchiral template application library that reduces incorrect template roundtrips by 90% relative to the original rdchiral and by 94% relative to rdchiral_cpp. Both libraries are freely available at https://github.com/denovochem/agave_chem and https://github.com/denovochem/rdchiral_plus.

Contents

1 Introduction

Atom-to-atom mapping (AAM) is the assignment of integer labels to atoms in a reaction SMILES [14] such that each product atom is paired with the reactant atom from which it originated. This correspondence underpins a broad range of computational chemistry tasks. Reaction template extraction—the automated derivation of SMARTS-based transformation rules from reaction databases—requires exact knowledge of which bonds are formed and broken, information that can only be obtained from a reliable atom map [5]. Retrosynthetic planning tools including ASKCOS [6, 19] and related systems consume these templates directly, so errors in the upstream map propagate throughout the synthesis prediction pipeline. Beyond template-based methods, atom maps also provide the training signal for sequence-to-sequence models of forward and retrosynthetic prediction [10] and are used in reaction classification workflows [11]. Despite its ubiquity as a dependency, producing accurate and complete atom maps at scale remains difficult.

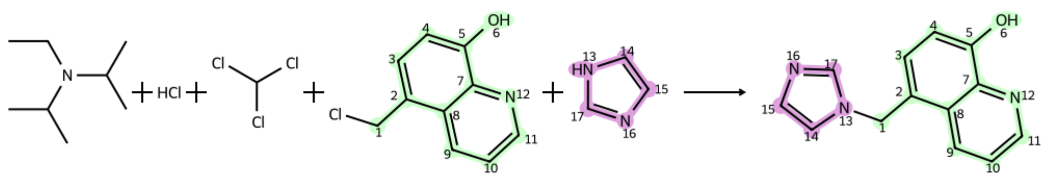

Figure 1: Example of atom-atom mapping (AAM) for a chemical reaction.

Graph-based approaches. Early mapping methods relied on heuristic graph matching. INDIGO and related tools use atom-invariant scoring and integer programming to find optimal correspondences [13]. While these approaches are interpretable and fast on small molecules, they do not generalize well to large or structurally complex reactions, and they have no mechanism to exploit the wealth of structural knowledge encoded in reaction databases. The benchmarking study of **(author?)** [7] systematically evaluated several classical and hybrid mappers and identified substantial failure rates on USPTO reactions, motivating the development of data-driven alternatives.

Template-based approaches. Another class of AAM approaches assigns atom correspondences by matching curated reaction SMIRKS or rule-based templates that encode the mechanistic transformation directly. NameRxn [12] (NextMove Software) classifies reactions against a large rule base of known mechanisms and named transformations, establishing atom-level correspondences as an inherent byproduct of the template match.

Neural approaches. Schwaller et al. [1] demonstrated that Transformer encoder models [17] trained on unlabeled reaction SMILES via masked language modeling (MLM) [18] learn atom-mapping as an emergent property of their attention weights, enabling attention-guided AAM without any labeled data. This unsupervised paradigm scales naturally to large reaction corpora and achieves strong overall accuracy, but the MLM objective has no mechanism to explicitly penalize incorrect correspondences; errors arise especially for symmetric substrates and reactions where attention provides insufficient discriminating signal. Nugmanov et al. [3] (GraphormerMapper) addressed the sensitivity of SMILES-based models to atom ordering by replacing sequential tokenization with a graph-structured Transformer, improving mapping stability and generalization across reaction types. Both of these approaches are trained entirely without labeled atom-mapping data.

A different strategy was taken by Chen et al. [2] (LocalMapper), who demonstrated that supervised training from chemist-verified reactions substantially improves AAM accuracy. Using a human-in-the-loop active learning framework, LocalMapper trains a graph neural network on a small set of manually annotated reactions and iteratively expands coverage by sampling uncertain predictions for additional human labeling. Starting from as few as 200 labeled reactions, LocalMapper achieves state-of-the-art accuracy on the USPTO-50K benchmark, with confident predictions showing 100% accuracy on 3,000 randomly sampled reactions [2]. However, per-reaction manual annotation is inherently expensive: even with active learning, the labeled set covers only ∼1,000 reactions across five labeling rounds, and accuracy on out-of-distribution reaction classes is bounded by the diversity of the annotated examples.

This work. AgaveChem takes direct inspiration from the observation that supervised training improves AAM accuracy [2], but replaces per-reaction annotation with template-level labeling. The key insight is that a curated reaction SMIRKS template encodes the atom correspondence for an entire reaction class: once a template has been verified, it generates correct ground truth for every reaction in the dataset that it matches—at no additional annotation cost. Composing template matching with an MCS-based partial mapper extends coverage to reactions without a matching template. Together, these deterministic methods produce a labeled training corpus from a filtered subset of the Lowe USPTO dataset that is orders of magnitude larger than what direct manual annotation can realistically achieve. The primary contribution of this work is a supervised ALBERT-based neural mapper trained on this labeled corpus. We show that direct supervision of the attention alignment objective substantially improves AAM accuracy relative to purely unsupervised training.

AgaveChem is released as an open-source Python library with four composable mappers:

(1) **Identical fragment mapper.** Fragments that appear structurally unchanged on both sides of the reaction arrow—counter ions, spectator molecules, and common leaving groups—are detected and assigned maps directly, before any other mapper is invoked.

(2) **MCS mapper.** Atoms whose local chemical environment is unchanged during the reaction are mapped via a modified MCS procedure. A configurable radius parameter controls how close to a reactive center the mapping extends, yielding conservative partial maps that avoid incorrectly assigning atoms near bond-breaking events.

(3) **Expert template mapper.** A curated library of reaction SMIRKS templates, sourced from ReactionFlash, Rxn-INSIGHT [4], and manual curation, is applied to the reaction product. If a template generates the correct reactants, the mapping encoded in the template SMIRKS is accepted as ground truth for that reaction.

(4) **Neural mapper.** An ALBERT transformer trained in two phases—unsupervised MLM pre-training followed by multitask supervised fine-tuning—that produces complete atom maps via attention alignment at inference time. This is the primary result of the paper.

The first three mappers serve a dual role: they generate the labeled training data for the neural mapper and are themselves released as independent library components. The ground truth generation pipeline is described in Section 3; it produces partial or complete atom maps for a filtered Lowe USPTO corpus, fully mapping ∼60% of reactions and covering ∼90% of product atoms.

As a secondary contribution, we release `rdchiral_plus`, a substantially refactored fork of rdchiral [5] that is used internally by the template mapper. It introduces deterministic stereochemistry handling, corrects bond direction artifacts in conjugated systems, supports one-pot template application, and reduces incorrect template roundtrips by 96% compared to the original rdchiral and 98% compared to rdchiral_cpp. Section 3.6 provides full details.

The paper is organized as follows. Section 2 reviews related work on atom mapping. Section 3 describes the AgaveChem system in full detail. Section 4 presents benchmarks against existing neural mappers. Section 5 discusses limitations and directions for future work.

2 Related Work

Expand related work section. Key papers to discuss: RXNMapper [1], LocalMapper [2], GraphormerMapper [3], Rxn-INSIGHT [4], rdchiral [5], and the benchmarking study [7]. Also cover NameRXN/NextMove and classical graph-matching approaches.

3 Methods

3.1 System Overview

AgaveChem is structured as a library of four independent mappers (Figure 2) that can be used individually or combined into a pipeline. In the default configuration, mappers are applied in order of increasing complexity and decreasing determinism: identical fragment → MCS → template → neural. Each mapper operates on unmapped reaction SMILES and returns a scored, mapped reaction SMILES together with metadata about how the mapping was produced. A higher-priority mapper in the pipeline can override the output of a lower-priority one. Users may substitute any subset of mappers or define custom selection strategies.

Insert system overview figure.

Figure 2: Overview of the AgaveChem mapping pipeline. The four mappers (identical fragment, template, MCS, neural) can be used independently or composed. In the ground truth generation phase, the first three deterministic mappers are used without the neural mapper; at inference time, the neural mapper is used as the primary mapper.

3.2 Dataset Preparation and Ground Truth Generation

Source data. We draw from the Lowe USPTO reaction dataset [9], which contains reactions extracted from US patent literature. We filter this dataset to remove duplicate reactions, reactions that could not be properly canonicalized, reactions that fully contain the product as a part of the reactants SMILES. We also devise a heuristic to remove reactions that erroneously contain a related, but unused chemical as a part of the reaction SMILES, typically as a result of a reference to a similar procedure.

total count

Insert reaction SMILES of erroneous reaction and accompanying passage from USPTO.

Describe filtering steps: duplicates removed, malformed SMILES removed, etc.

After filtering, approximately 0.97 million reactions remain. Atom mapping annotations present in the original dataset are stripped prior to any processing so that the ground truth is generated entirely by the pipeline described below.

Ground truth generation pipeline. The three deterministic mappers are applied sequentially to each unmapped reaction. A reaction is considered fully mapped if every product atom receives a mapping to a unique reactant atom. Partial maps (from the MCS mapper) are retained for reactions that are not fully covered by the template mapper. Table 1 summarizes the coverage achieved at each stage.

Table 1: Ground truth coverage at each stage of the deterministic mapping pipeline across the ∼0.97M filtered USPTO reactions.

Stage	**Fully mapped (%)**	**Product atoms mapped (%)**
Identical fragment mapper only	*TBD*	*TBD*
+ MCS mapper	*TBD*	*TBD*
+ Template mapper	∼60	∼90

3.2.1 Identical Fragment Mapper

The identical fragment mapper handles the simplest class of atom-mapping decisions: fragments that appear structurally unchanged on both sides of the reaction arrow. These include counter-ions, solvent molecules, spectator reagents, and any other disconnected components whose canonical SMILES is identical in both the reactant and product.

Algorithm. Each reactant and product component is first independently canonicalized with RDKit [15]. A fragment is identified as *identical* if its canonical SMILES appears in both the reactant component list and the product component list. For each such fragment, every atom is assigned a unique atom-map number (allocated from a reserved range starting at 500 to avoid collisions with downstream mappers), and the fragment is removed from both sides of the reaction before any other mapper is invoked. After downstream processing of the reduced reaction, the pre-mapped identical fragments are reinserted on both sides, yielding a complete reaction SMILES in which all spectator atoms carry consistent map numbers.

This design serves a dual purpose. First, identical spectators are handled correctly before they can confuse downstream mappers by introducing spurious reactant–product correspondences. Second, for reactions in which the only unmapped components are identical spectators, the mapper provides a complete map without invoking the more expensive template or MCS procedures.

3.2.2 MCS Mapper

The MCS mapper assigns atom-map numbers to atoms whose local chemical environment is preserved across the reaction. Rather than solving a full maximum common substructure (MCS) problem—which is NP-hard in general and scales poorly to multi-fragment reactions—it uses a bond-radius environment fingerprinting scheme that identifies invariant atoms efficiently and naturally parameterizes how close to the reactive center the mapping is permitted to extend.

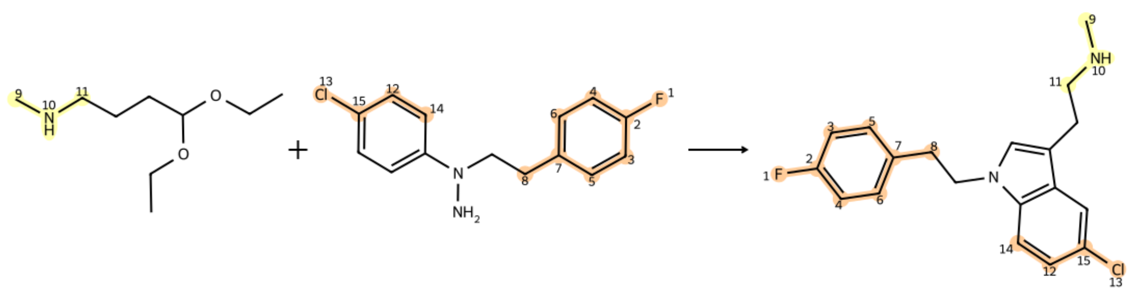

Figure 3: Example of partial atom-atom mapping (AAM) for a chemical reaction using the MCS mapper.

Environment fingerprints. For each atom a in a molecule and each integer radius r, the mapper constructs an *environment skeleton*: the canonical ordered list of bonds lying within r bonds of a, extracted via RDKit's `FindAtomEnvironmentOfRadiusN` and sorted by RDKit canonical atom ranks. From this skeleton a hashable, index-free *fingerprint* is computed by encoding, for each bond in the skeleton, the BFS distance from a to each bond endpoint together with the chemical properties of those endpoints (atomic number, formal charge, aromaticity, ring membership, total hydrogen count, and degree) and the bond itself (bond order, stereo). Two environments are considered identical if and only if their fingerprints match. The fingerprint also incorporates current atom-map numbers, so environments containing already-mapped atoms acquire a distinct signature that anchors subsequent matching.

Prior to fingerprint computation, each molecule is normalized by neutralizing formal charges and canonicalizing its tautomeric form via RDKit's `Uncharger` and `TautomerEnumerator`. This prevents charge variants and tautomers from generating spurious fingerprint mismatches for otherwise equivalent atoms.

Radius search. Fingerprints are built incrementally, starting from a minimum radius (default 1) and increasing until a unique reactant–product fingerprint match is found or all matches disappear. Atoms whose environment has no cross-side match at a given radius are pruned and skipped at all subsequent radii, keeping computation tractable. The largest radius at which at least one match was found is used as the *final radius* for the subsequent mapping phase. Fingerprint lookup uses a hash-based inverted index, giving $O(R + P)$ complexity per radius rather than $O(R \times P)$ pairwise comparison.

Anchor-extend assignment. Once the final radius is determined, atom-map numbers are assigned via an *anchor-extend* strategy alternating between two phases:

(1) **Extend phase.** Scanning from the highest radius down to radius 1, the mapper identifies reactant–product fingerprint pairs that contain at least one already-mapped atom (*anchored* matches). Because the fingerprint includes existing map numbers, an anchored match is unambiguous: the matched atom is adjacent to a confirmed correspondence and can be safely propagated. The full sweep repeats until no further anchored matches remain.

(2) **New-anchor phase.** A single unanchored match is selected at the highest available radius (at or above `min_radius_to_anchor_new_mapping`, default 3) and assigned, then control returns to the extend phase.

This ordering ensures that each anchor is fully propagated before a new independent anchor is introduced, preventing inconsistent mapping chains. A candidate assignment is additionally rejected if the product atom has an already-mapped neighbor that was sourced from a different reactant molecule, enforcing structural adjacency consistency.

Conservative partial mapping. The `min_radius_to_anchor_new_mapping` parameter controls conservatism near the reactive center. At low radii, local environments are too generic to uniquely distinguish atoms that are unchanged from those involved in bond-making or bond-breaking events. By requiring new unanchored matches to carry at least radius 3 of local context, the mapper abstains from assigning atoms in ambiguous positions. The result is a conservative *partial* map: atoms well removed from the reaction site receive confident assignments, while atoms near the reactive center are left unmapped for the neural mapper to resolve.

3.2.3 Expert Template Mapper

The template mapper applies a curated library of reaction SMIRKS to each reaction product and accepts a mapping when the virtual reactants generated by the template match the actual reactants after canonicalization. Because the template encodes the mechanistic atom correspondence explicitly, every accepted map correctly captures bond-breaking and bond-forming events—not merely the invariant substructure. This makes template-produced maps the highest-quality ground truth in the pipeline.

Template library. Templates were assembled from three sources: (1) the Rxn-INSIGHT library of named reaction patterns [4]; (2) templates extracted from the ReactionFlash iOS application; and (3) a set of nucleophilic substitution templates generated programmatically. All templates were reviewed and, where necessary, corrected by hand. Each parent SMIRKS is expanded into one or more *child* SMIRKS that capture specific substitution patterns or stereo-variants of the parent; templates are applied at the child level. Each template carries an optional two-level priority tuple `(priority_class, priority)` used for tie-breaking when multiple templates match a single reaction. Template application is performed using `rdchiral_plus` (Section 3.6).

MCS pre-pass. Before any template is applied, the MCS mapper is run on the input reaction to produce a partial map. Connected components of product atoms that the MCS mapper could not assign—atoms near the reactive center whose local environment is not uniquely shared with a reactant atom—are identified as *unmapped islands*. Template application is then restricted to templates whose product SMARTS overlaps with at least one unmapped island. This focuses the template search on the reaction center and avoids wasting time on templates that would only remap atoms already conservatively assigned by the MCS mapper.

Two-tier template screening. For each template, candidate viability is assessed in two stages before the full rdchiral application is invoked:

(1) **Fingerprint pre-screen.** RDKit `PatternFingerprint` vectors are pre-computed for every template fragment and for every product and reactant molecule. A template is discarded immediately if any of its required product or reactant bits are absent from the corresponding molecular fingerprints (`AllProbeBitsMatch`). This bitwise check eliminates the vast majority of templates in $O(1)$ per template, before any substructure search is performed.

(2) **Substructure pre-screen.** For templates that pass the fingerprint check, each template fragment must have at least one substructure match in the product molecules (and, independently, in the reactant molecules). When unmapped islands are present, the product check is further tightened: a template fragment is accepted only if its substructure match has at least partial overlap with an unmapped island.

Only templates passing both screens are forwarded to full rdchiral application.

Template application and outcome validation. Templates passing both screens are applied to the product using `rdchiralRun`, which returns virtual reactant SMILES together with the atom map indices of the affected atoms. Each outcome is then validated:

(1) **Fragment matching.** The virtual reactant fragments are compared against the actual reactants (and their tautomers, enumerated once per reaction). A fragment is considered matched if its canonical SMILES—with atom map numbers stripped—appears in the tautomer set of any actual reactant.

(2) **Wildcard fragment resolution.** Templates with wildcard atoms (*) produce fragments whose canonical form does not directly match an actual reactant. For these, the mapper attempts to transfer the template's atom-map numbers to the actual reactant via substructure matching. The transfer is accepted only when the match is unique or when all matches are symmetry-equivalent (i.e. the molecule is symmetric with respect to the query), preventing ambiguous assignments.

(3) **Spectator reconciliation.** Any actual reactant component not present in the template outcome is reattached as an unmapped spectator so that the final reaction SMILES accounts for all input fragments.

An outcome is rejected if any fragment cannot be matched or resolved.

Outcome selection. Valid outcomes are canonicalized and deduplicated. Each candidate is cross-checked by stripping its atom-map numbers and verifying that the resulting reaction SMILES is identical to the canonicalized input reaction, ensuring the template did not introduce extraneous or missing atoms. If multiple distinct valid outcomes remain after deduplication, the preferred mapping is selected first by template priority class, then by intra-class priority, and finally—when no explicit priority is set—by the total atom count of the matching template (more structurally specific templates are preferred).

3.3 Neural Mapper Architecture

The neural mapper follows the same general architecture as RXNMapper [1]: an ALBERT transformer encoder whose attention weights in a designated head and layer are used to produce atom-to-atom correspondences. The model uses the same ALBERT configuration as RXNMapper (hidden size 256, 12 layers, 8 attention heads, intermediate size 512, embedding size 128), but with a custom vocabulary of size 1024 tokens designed specifically for reaction SMILES. Reactions are tokenized at the atom level using a custom tokenizer that treats each SMILES atom token (including bracket atoms and isotope specifications) as a single unit, preserving bond, ring closure, and branch tokens as separate tokens.

For supervised training, we augment the base ALBERT model with a dedicated attention alignment head. The alignment head extracts the raw attention logits from a specific layer and head identified during the unsupervised pre-training phase, applies a final linear projection, and computes a cross-entropy loss against the ground truth alignment matrix. Unmapped atoms and non-atom tokens (bonds, parentheses, ring-closure digits) are directed to a special "sink" token, making the loss well-defined even for partially labeled reactions.

Symmetric atoms—those that belong to symmetry-equivalent groups as determined by RDKit's canonical ranking—receive smoothed targets that distribute probability uniformly over the symmetric group, preventing the model from being penalized for correct but arbitrarily labeled assignments.

Insert model architecture figure: ALBERT encoder, attention alignment head, supervised loss.

Figure 4: AgaveChem neural mapper architecture. A custom attention alignment head extracts logits from a designated attention head and computes a cross-entropy loss against the ground truth alignment matrix. Unmapped atoms are directed to a sink token. Symmetric atoms receive smoothed targets.

3.4 Training Procedure

Training proceeds in two phases.

3.4.1 Phase 1: Unsupervised MLM Pre-training

In the first phase, the base ALBERT model is trained on the full corpus of filtered USPTO reactions using masked language modeling. We adopt a graph-aware span masking strategy in place of the standard independent token masking used by RXNMapper. Rather than selecting individual tokens uniformly at random, the masking procedure selects contiguous neighborhoods of atoms on the molecular graph, using BFS from a randomly chosen seed atom with a stochastic span size drawn from a weighted distribution over $\{1, 2, 3, 4, 5\}$ atoms. Only atom tokens are eligible for masking; structural tokens (bond symbols, parentheses, ring-closure digits) are never masked. For each masked atom, the replacement follows a modified 70/20/10 strategy: 70% of the time the token is replaced with `[MASK]`, 20% of the time it is replaced with a chemically plausible alternative (drawn from curated bioisosteric substitution groups), and 10% of the time the original token is retained. Random SMILES augmentation is applied at each epoch: with 5% probability the canonical form is used; otherwise, a random SMILES enumeration of the reaction is sampled, with a 10% probability of also randomizing the tautomer.

The model is trained with AdamW with a learning rate of 2×10^{-4}, weight decay 10^{-3}, linear warmup over 10,000 steps, and gradient clipping at 1.0. The masking probability is 15% in this phase.

3.4.2 Phase 2: Multitask Supervised Fine-tuning

In the second phase, training continues on all reactions for which ground truth atom maps are available—both fully and partially mapped by the deterministic pipeline. Fully mapped reactions ($\sim$60% of the labeled corpus) contribute supervision over all product atoms; partially mapped reactions contribute supervision over only the subset of atoms that received a ground truth assignment. The training objective is a weighted combination of the MLM loss and the supervised attention alignment loss:

$$\mathcal{L} = \lambda_{\text{MLM}} \cdot \mathcal{L}_{\text{MLM}} + \lambda_{\text{align}} \cdot \mathcal{L}_{\text{align}}, \tag{1}$$

where $\mathcal{L}_{\text{align}}$ is the per-atom cross-entropy between the predicted attention distribution and the ground truth alignment matrix, computed only over atoms for which a ground truth mapping is available. Continuing the MLM objective prevents catastrophic forgetting and encourages the model to maintain the chemical representation learned in Phase 1. The masking probability is increased to 20% in this phase, and span masking uses the same weighted distribution over span sizes.

Report final λ values and other hyperparameters after ablation experiments.

3.5 Inference

Describe the inference procedure: how attention weights are extracted, how correspondences are decoded from the attention matrix (argmax, Hungarian algorithm, etc.), and how the mapping is converted back to atom map numbers in the SMILES string.

3.6 `rdchiral_plus`

The expert template mapper relies on accurate and reproducible template application. The original rdchiral library [5], while widely used, contains several classes of errors that affect both template extraction and application. We developed `rdchiral_plus`, a fork of rdchiral with the following improvements.

Template application.

- *Conjugated system bond direction correction.* Corrupted single-bond directions (`ENDUPRIGHT`, `ENDDOWNRIGHT`) that arise in conjugated systems are detected and corrected.
- *Broader stereochemistry handling.* Tetrahedral centers with lone pairs are now handled correctly.
- *One-pot reactions.* Templates defining multiple transformations on the same product are initialized with parentheses, enabling correct handling of multi-site reactions.
- *Recursive template application.* Templates can be applied recursively with a configurable maximum depth, which is necessary for symmetric reactions or reactions occurring at multiple sites.

Template extraction.

- *Configurable radius.* Template extraction supports a configurable radius and special group handling.

- *Deterministic extraction.* The random shuffle-based tetrahedral correction loop in the original implementation is replaced with a deterministic permutation parity calculation, eliminating rare hangs and non-reproducible outputs.
- *Stereochemistry tracking.* Inversion of tetrahedral centers is now counted as a changed atom and included in extracted templates.
- *Spectator tracking.* Spectator molecules are included in extracted template dictionaries.

Consistency with rdchiral. Roundtrip consistency—extracting a template from a mapped reaction SMILES and re-applying it to the product to recover the expected reactants—is used to quantify correctness. `rdchiral_plus` reduces incorrect roundtrips by 90% compared to rdchiral and by 94% compared to rdchiral_cpp. Table 2 summarizes the comparison.

Table 2: Roundtrip consistency of template extraction and application across the three rdchiral variants. A roundtrip is considered correct if re-applying the extracted template to the product recovers the expected reactant SMILES.

Library	**Incorrect roundtrips (%)**	**Reduction vs. rdchiral_plus**
rdchiral	*TBD*	−96%
rdchiral_cpp	*TBD*	−98%
rdchiral_plus	*TBD*	—

4 Results and Discussion

We evaluate AgaveChem against three established neural mappers—RXNMapper [1], LocalMapper [2], and GraphormerMapper [3]—using the benchmark protocol and evaluation sets described in [7, 16], employing held-out reactions drawn from the USPTO and graph isomorphism to determine whether a proposed mapping is equivalent to the reference. All reactions included in any evaluation set were withheld from AgaveChem's training corpus prior to training. We report two primary metrics: *exact match accuracy*, defined as the fraction of reactions for which every product atom is correctly mapped to its reference counterpart, and *per-atom accuracy*, defined as the fraction of individual product atoms across all test reactions that receive the correct mapping. Exact match is the more stringent criterion; per-atom accuracy better characterizes performance on partial or near-correct mappings and is informative for reactions where the model assigns only a small number of atoms incorrectly.

4.1 Benchmark Datasets and Evaluation Metrics

Describe the benchmark dataset(s) in detail: source, size, reaction types covered, split strategy. Reference the benchmarking study of Jaworski et al. and any additional evaluation sets used.

4.2 Comparison with Existing Neural Mappers

Table 3: Atom mapping accuracy on the benchmark test set. Exact match: fraction of reactions with all atoms correctly mapped. Per-atom: fraction of individual product atoms correctly mapped. All AgaveChem results use the neural mapper.

Method	**Exact match (%)**	**Per-atom (%)**
RXNMapper [1]	*TBD*	*TBD*
LocalMapper [2]	*TBD*	*TBD*
GraphormerMapper [3]	*TBD*	*TBD*
AgaveChem (neural)	***TBD***	***TBD***

Write the body of this subsection once benchmark numbers are available. Key points: (1) AgaveChem outperforms LocalMapper and substantially outperforms RXNMapper; (2) characterize where remaining errors occur; (3) include qualitative examples.

4.3 Template Mapper Performance

Report template mapper performance on named reactions: exact match rate for template-covered reactions, coverage across reaction classes, comparison to rule-based mappers.

4.4 Ground Truth Quality Analysis

Analyze the quality of the generated ground truth: sample a subset of fully mapped reactions and verify against manual annotation or alternative references; characterize which reaction classes the deterministic pipeline fails to map.

5 Conclusions

We have presented AgaveChem, an open-source atom-to-atom mapping library that combines deterministic expert knowledge with supervised neural learning. The library provides four composable mappers—identical fragment, expert template, MCS, and neural—each targeting a different subset of the atom mapping problem and suited to different downstream use cases.

The central contribution is the supervised training strategy for the neural mapper. Rather than relying solely on unsupervised MLM, we use the deterministic mappers to generate labeled ground truth from a filtered Lowe USPTO corpus, then apply this data to directly supervise the attention alignment objective during fine-tuning. The result is a model that retains the broad reaction coverage of attention-based neural mappers while substantially reducing the class of errors that arise from inferring atomic correspondence without explicit supervision. The deterministic mappers that produce the training labels are themselves useful tools, covering ∼60% of reactions fully and ∼90% of product atoms across the USPTO corpus.

The expert template mapper is notable as a standalone tool. For reactions that match a known named reaction pattern, it produces maps that are by construction mechanistically consistent, making it particularly well-suited for downstream applications that require interpretable maps, such as reaction classification and retrosynthetic template extraction.

As a byproduct of this work, we release `rdchiral_plus`, a substantially improved fork of rdchiral. The 90% reduction in incorrect template roundtrips directly improves the quality of the generated ground truth and, through it, the quality of the trained neural model.

Limitations and future work. The ground truth generation pipeline produces only partial maps for roughly 40% of training reactions, leaving the neural model to generalize over reaction types that are underrepresented in the labeled data. While the model already outperforms existing baselines, performance on unusual reaction classes and highly symmetric substrates remains an area for improvement.

The pipeline is well-suited to incremental improvement: expanding the expert template library—particularly for less common named reactions and multi-step transformations—directly increases the fraction of fully labeled training reactions at no additional per-reaction annotation cost. More broadly, any deterministic method that can precisely establish product atom provenance—additional template sources, curated mechanistic rules, or reaction-class-specific substructure procedures—slots naturally into the same pipeline. Self-training, in which high-confidence predictions from the neural mapper are added back to the labeled set, may further close the accuracy gap without requiring additional annotation. Training on a larger, higher-quality, and more chemically diverse corpus would likewise be expected to improve generalization, especially for underrepresented or complex reaction classes.

The labeled ground truth produced by the deterministic pipeline is architecture-agnostic. While the current neural mapper uses an ALBERT-based SMILES encoder, the same supervision signal could be applied to graph neural network architectures, alternative transformer models, or other sequence encoders that operate on reaction SMILES. No systematic hyperparameter search or architecture ablation was conducted in this work; exploring these dimensions represents a natural direction for future investigation.

Further improvements to AgaveChem app - reranking, scoring, extensible mapping.

Acknowledgements

Add acknowledgements.

Data and Software Availability

AgaveChem is released as an open-source Python library under the MIT license at https://github.com/denovochem/agave_chem. `rdchiral_plus` is available at https://github.com/denovochem/rdchiral_plus.

Add data availability statement for the training set and benchmark reactions.

References

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